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Leveraging ChatGPT for Second Language Writing Feedback and Assessment

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ABSTRACT

This paper discusses the increasingly prominent role of ChatGPT in providing feedback and assessment for L2 writing in the digital age. It reviews representative studies that address five research strands about the use of ChatGPT in L2 writing contexts. After a critical evaluation of the existing literature, the author extensively explains four innovative sub-topics on leveraging ChatGPT for L2 writing feedback and assessment, drawing on multiple creative projects undertaken by her research team. These include: 1) ChatGPT-student collaboration in the L2 writing process; 2) ChatGPT-supported teacher feedback; 3) collaborative processing of ChatGPT feedback; and 4) the potential of ChatGPT for L2 writing assessment. This paper also addresses the importance of teacher training on the use of GenAI in language and writing instruction.

KEYWORDS

Generative AI, ChatGPT, Second Language Writing, Writing Feedback, Writing Assessment

INTRODUCTION

This paper explores the role of the Generative Artificial Intelligence (GenAI) tool ChatGPT in second language (L2) writing. It begins with a brief review of the current literature, highlighting five research strands about the use of ChatGPT in L2 writing contexts. Building on existing research, the author presents four innovative sub-areas developed with her team, aiming to provide valuable insights and foster discussion on leveraging AI in instruction for L2 researchers and teachers.

GenAI is defined as a type of machine learning that uses AI to create new content, including text, images, videos, and audio. It is characterized by its human-like conversational abilities and capacity for knowledge expression. GenAI has been widely utilized in our daily life, and some GenAI tools have been increasingly applied in education, including text-based AI (e.g., ChatGPT, Gemini, Wenxinyiyan, iFLYTEK), text to image AI (e.g., Dall.E and Midjourney), text to video AI (e.g., Fliki, Synthesia), and text to presentation AI (e.g., Gamma, PopAi). To deepen our understanding of the affordances and constraints of AI in L2 teaching and learning, several top-tier journals have called for collective scholarly insights through special issues (SIs). *Language Learning & Technology* has published a SI on “Artificial intelligence for language learning: Entering a new era” (Warschauer & Xu, 2024). This SI explored diverse applications of AI tools in practicing input skills (e.g., listening) and output skills (i.e., speaking and writing), examining both outcomes and processes from the perspectives of students and teachers. Focusing on the role of GenAI in writing, *Computers and Composition* has published a SI entitled “Composing with generative AI”, which examined the use of GenAI tools “as assistants to human-driven writing tasks” (Ranade & Eyman, 2024, p. 3). With the aim to exploring

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the effects of GenAI on language teaching and learning, the SIs of *TESOL Quarterly* (TQ) and *System* are coming out in 2025. TQ will explore using GenAI for teaching, learning, and assessing English productive skills, while *System* will delve into opportunities, challenges, and ethical considerations regarding using GenAI for language pedagogy. Moreover, *Journal of Second Language Writing* will publish a SI on "Generative artificial intelligence and L2 writing" (2025) and *Language Testing* also has a SI ongoing entitled "Advancing language assessment for teaching and learning in the era of AI revolution: Promises and challenges" (2025). A surge of research will follow the publication of these aforementioned SIs in 2025.

As a prominent GenAI tool, ChatGPT has gained recognition as a large language model (LLM) capable of producing customized output, and as a fast, versatile conversational agent. The free version of ChatGPT (i.e., ChatGPT-3.5, now known as ChatGPT 4o) has been particularly beneficial to L2 teachers and students worldwide. Previous research has extensively reported on the use of ChatGPT in L2 instruction. This paper specifically focuses on its application in L2 writing and discusses five research strands identified through a review of literature, with a few representative studies presented. Next, the author explains four innovative subtopics, drawing on her recent creative research projects.

FOCUSED LITERATURE REVIEW

Many articles examining the use of ChatGPT in L2 writing were published during 2023 and 2024. The first research strand focuses on the impact of ChatGPT on L2 writing development (e.g., Mahapatra, 2024; Strobl et al., 2023). For instance, Strobl et al. (2023) explored how ChatGPT influenced the revision process of advanced L2 German writers at a Dutch university. Students were tasked with assessing the quality of their own synthesis texts compared to those generated by ChatGPT. Afterwards, they revised their own texts. The findings revealed that the students rated their own texts as lower in quality than ChatGPT's. They made positive comments on academic language use, varied lexicon, organization, and content in ChatGPT-generated texts.

Connected to this is the second strand, which compares ChatGPT-generated essays with human-written ones (e.g., Nkhobo & Chaka, 2023; Zindela, 2023). For example, Zindela (2023) compared 20 student essays written by Tswana learners of English in Africa with essays produced by ChatGPT 3.5 on the same topics. The comparison showed some differences: L2 students produced longer sentences while ChatGPT used more sophisticated and varied vocabulary. In contrast, students relied more on function words and had less lexical variety. Interestingly, in Strobl et al.'s (2023) study, students made numerous content and language revisions based on ChatGPT's texts, some of which were over-revisions. This observation points to the challenge in students' use of ChatGPT output. It underscores the need for students to take ownership of their writing while collaborating with ChatGPT. The first innovative research topic discussed in the following section *student-ChatGPT collaboration* during research writing, addresses such a challenge and advocates for greater learner agency and autonomy.

The third strand focuses on ChatGPT feedback on L2 writing, in comparison to teacher feedback (e.g., Fokides & Peristeraki, 2024; Guo & Wang, 2024). A representative study by Guo and Wang (2024) compared the feedback provided by teachers and by ChatGPT on L2 writing. In this study, 5 Chinese EFL teachers graded argumentative essays written by 50 undergraduate EFL students, which were also graded by ChatGPT-3.5. Afterwards, the teachers reviewed the feedback generated by ChatGPT and shared their perceptions through individual interviews. The results revealed that ChatGPT produced significantly more feedback than the teachers. Teacher feedback tended to focus on content and language issues, while ChatGPT offered more balanced attention to content, organization and language. From the teachers' perspectives, GPT is more powerful than existing automated writing evaluation (AWE) tools, with the potential to reduce their workload and improve their feedback literacy. However, they noted that ChatGPT lacks an understanding of the specific curriculum, students' language proficiency, and individual personalities, leading to occasional inappropriate feedback. This study highlights the need for collaboration of ChatGPT and teachers in providing writing feedback. The second innovative research topic *ChatGPT-supported teacher feedback*, to be discussed in the next section, well addresses the limitations of ChatGPT in L2 writing feedback and creates a synergy between ChatGPT's feedback and the teacher's professional judgment.

The fourth strand is ChatGPT assessment of L2 writing in comparison to human raters (e.g., Geçkin, Kızıltaş & Çınar, 2023; Jiang et al., 2023; Mizumoto et al., 2024; Pfau & Matthew, 2023). For example, Pfau and Matthew (2023) compared ChatGPT's evaluation of accuracy with human coding across various proficiency levels, particularly its consistency in identifying errors. In their study, both teachers and ChatGPT assessed 100 essays from a corpus of writing by Greek learners of English. The findings revealed a strong correlation between ChatGPT's error detection and human coding. Nevertheless, ChatGPT struggled to consistently identify language errors in low-level essays. The use of ChatGPT in L2 assessment is a growing trend, and while accuracy is important, future research should explore its potential in assessing multiple writing dimensions beyond error detection using comprehensive grading rubrics. The fourth innovative research topic *the potential of ChatGPT for L2 writing assessment* will address ChatGPT's ability to assess multiple writing areas.

The fifth strand examines learners' and teachers' perceptions on ChatGPT's role in L2 writing (e.g., Bao & Li, 2023; Yan, 2023; Yuan et al., 2024; Zou & Huang, 2023). Yuan et al.'s (2024) study presents a representative example. Seven Chinese EFL students participated in the study and shared their perceptions on using ChatGPT for academic writing. They praised ChatGPT as a revolutionary tool --convenient and responsive to their needs, particularly for scaffolding their writing in terms of content and fluency. However, they expressed concerns about the authenticity of information generated, as well as the risks of academic plagiarism and a lack of critical thinking. Effectively and appropriately using ChatGPT for academic writing warrants further exploration. Instead of individual interaction, students working in pairs or small groups can engage with ChatGPT outputs collectively. Through collective scaffolding, they are more likely to utilize ChatGPT successfully. The section below includes an innovative research topic on *collaborative processing of ChatGPT feedback*.

INNOVATIVE RESEARCH TOPICS

This section presents four innovative research topics addressing ChatGPT for L2 writing feedback and assessment. For each topic, the author refers to the empirical study that her research team have conducted[1].

ChatGPT-Student Collaboration in L2 Writing

Previous research has shown that ChatGPT helps students brainstorm, organize ideas, and address both global and local writing issues in L2 writing (Wang, 2024). Similarly, ChatGPT can assist with topic elucidation, structuring, feedback rendering, and linguistic refinement (Barrot, 2023). Warschauer et al. (2023) suggest that AI tools can significantly reduce the linguistic challenges faced by L2 writers, potentially serving as equalizers in addressing linguistic disparities. ChatGPT can be a valuable companion for L2 students in the writing process, as they consistently engage in learning while interacting with ChatGPT for constructive feedback.

The author's team (Cheng, Li, and Lee) examined the role of ChatGPT in supporting ESL graduate students with research writing. The focal cases were two first-year international doctoral students in an Ed.D. program in Higher Education at a public university in US, who were working on a course research project exploring the history of a significant educational topic. The research team developed a comprehensive PowerPoint tutorial and trained them on how to use ChatGPT for research writing. The training was informed by the AI literacy framework proposed by Warschauer et al. (2023), as outlined in Table 1 below.

The analyses of the archived ChatGPT records of the two doctoral students Sam and Ian[2], including the prompts they input, ChatGPT's output, and transcripts from semi-structured individual

Table 1. AI literacy framework (adapted from Warschauer et al., 2023)

Step	Description
Step 1 Understanding AI tools	Introducing features and functions of ChatGPT
Step 2 Utilizing AI for specific tasks	Providing examples demonstrating how to effectively use ChatGPT for research writing tasks
Step 3 Skillfully prompting AI	Emphasizing the importance of writing clear prompts and including strategies for hypothesizing responses and adapting questions when needed.
Step 4 Verifying AI-generated content	Teaching how to evaluate the accuracy and reliability of AI-generated content by cross-verifying with other resources
Step 5 Ethically integrating AI output into writing	Addressing ethical considerations and avoiding plagiarism.

Table 2. Students' inquiries about their research writing

Category	Example
Content	"Provide me with a few research topic options on the history of technology in higher education".
Clarity	"Can you check the paragraph and see if that is making sense?"
Coherence	"Are there any instances of coherence breaks in the following text, such as misleading disposition, misleading heading, irrelevance, unfocused paragraph, and lack of transitions?"
Documentation	"Revise the references following the APA style."
Genre	"How to write a good literature review?"
Language use	"Check for grammatical errors please."

Table 3. Two doctoral students' ChatGPT inquiries and adoption of ChatGPT feedback

	Student	Content	Clarity	Coherence	Documentation	Genre	Language use	Total
ChatGPT inquiries	Ian	5	2	2	2	6	4	21
	Sam	21	25	29	49	6	10	140
Adoption of feedback	Ian	60%	50%	50%	100%	66.7%	25%	57%
	Sam	12.9%	88%	93.1%	85.7%	66.7%	80%	80%

interviews, revealed that the students consulted ChatGPT-3.5 on various aspects of their research writing, such as content, clarity, coherence, documentation, and language use. Table 2 provides an illustrative excerpt for each category. Interestingly, the two students demonstrated very different approaches to using ChatGPT for their research writing, as shown in Table 3. Sam engaged actively with ChatGPT, submitting a total of 140 inquiries, especially focused on documentation. Ian, on the other hand, was more cautious, making only 21 inquiries. Regarding their revisions, Sam incorporated the majority of ChatGPT's feedback, except for content, in which he adopted merely 12.9%. Ian, in contrast, incorporated around half of ChatGPT's output, fully adopting its feedback related to documentation (100%).

In the individual interviews, both students highlighted the advantages of using ChatGPT, such as time efficiency, knowledge clarification, and affective benefit. Sam viewed ChatGPT as a writing tutor, a companion (as opposed to the instructor), and a valuable learning tool. For instance, Sam commented that "It really felt it is like a tutor, you know, like it's just someone there to help me understand the key terminology." and "I don't need to worry about losing face in front of a human instructor." However, Ian expressed more concerns about the potential negative effects of using

ChatGPT, inferring the risk of academic plagiarism. For example, he stated "You have to use it in a way that does not harm you. Yeah, you cannot rely on it 100%. No, not at all."

Research on ChatGPT-student collaboration in L2 writing is still in its infancy stage. Future studies should explore this topic in diverse writing contexts (e.g., English for specific purposes) and writing tasks, including more informal writing like email construction. Triangulated data sources, such as weekly reports, post-task interviews, surveys, and AI archives, along with the techniques like stimulated recall can provide deeper insights. A notable gap in the field is a lack of longitudinal experimental studies that investigate the long-term effects of AI tool usage on students' academic writing development.

ChatGPT-Supported Teacher Feedback

Previous research has pointed out the importance of collaboration between ChatGPT and teachers in providing L2 writing feedback (e.g., Guo & Wang, 2023). To enhance this collaboration, the author's team (Han & Li, 2024) proposed ChatGPT-supported teacher feedback, which refers to teachers providing L2 writing feedback based on ChatGPT output. This approach is gaining traction, due to strong pedagogical rationales. In many EFL contexts, large class sizes and heavy teaching loads limit the time instructors can dedicate to detailed written feedback. Using ChatGPT as an initial step can reduce teachers' workload while enabling them to offer high quality feedback. Notably, ChatGPT delivers personalized and comprehensive feedback that goes beyond language errors. However, since ChatGPT can generate inaccurate or unclear responses and lacks the ability to tailor content for specific audiences (Barrot, 2023), the teacher's role remains crucial.

The sample study (Han & Li, 2024) explores the efficacy of ChatGPT-supported teacher feedback. The participants included four EFL instructors and their undergraduate students majoring in World Language Education at a large public university in China. The students completed two writing tasks: an argumentative essay and an expository essay. ChatGPT were trained to respond to two prompts: 1) Prompt #1: "Please point out all errors on sentence structure, word choice, verb tense, noun endings (singular/plural), verb form, punctuation, articles/determiners, word form, spelling, run-ons, pronouns, subject–verb agreement, fragments, idioms, and informal text"; 2) Prompt #2: "Please provide feedback on the rhetoric within the article". For teacher training, the four instructors first reviewed the ChatGPT-generated feedback on a few sample essays, and then provided indirect coded feedback (corrective feedback) [3] with the research team's guidance. They also evaluated ChatGPT's rhetorical feedback [4] and made necessary modifications for the personalized holistic feedback, with the research team's guidance.

The results showed five types of adaptations that teachers made to ChatGPT-generated feedback, namely addition, extension, modification, condensing, and deletion. Table 4 below includes illustrative excerpts showing the justifications for each type of adaptation, derived from interviews with the instructors. Furthermore, the students effectively utilized ChatGPT-supported teacher feedback during their writing revisions. Table 5 below shows how this new feedback approach helped students improve their essays, with a high percentage of successful revisions in response to corrective feedback and a decent percentage of comprehensive modifications addressing rhetorical feedback.

This study shows that ChatGPT has great potential to assist teachers in delivering high-quality, personalized feedback efficiently, tailored to their pedagogical needs. Future research could explore this topic further by examining diverse L2 writing contexts and writing tasks. Additionally, it would be valuable to investigate both students' and instructors' perspectives on ChatGPT-supported teacher feedback. Comparative studies that analyzed similarities and differences between ChatGPT-supported teacher feedback and traditional teacher feedback would offer important insights. Furthermore, experimental studies need to examine the impact of ChatGPT-supported teacher feedback on long-term writing development, in comparison to traditional teacher feedback and Automated Writing Evaluation (AWE).

Table 4. Excerpts explaining instructors' adaptation of ChatGPT feedback

Adaptation type	Justification
Addition	"ChatGPT is unable to identify all mistakes, so I scan through the compositions to spot any remaining errors."
Extension	"I then provide a consolidated summary of recurring mistakes and grammatical errors observed in their compositions."
Modification	"I slightly modify the feedback to reduce the level of praise."
Condensing	"It is necessary to condense these for time efficiency and clear understanding for the students".
Deletion	"If I find that ChatGPT has modified such expressions [translation of Chinese idioms], I may choose to delete the [ChatGPT] modification without notifying the students."

Table 5. Students' incorporation of ChatGPT-supported teacher feedback (adapted from Han & Li, 2024)

	Task 1 (percentage)	Task 2 (percentage)
Incorporation of corrective feedback		
No change	23.6%	20.3%
Unsuccessful revision	11.9%	15.8%
Successful revision	64.5%	63.9%
Incorporation of rhetorical feedback		
No change	65.3%	35.8%
Partial modification	25.3%	7.4%
Comprehensive modification	9.5%	56.8%

Collaborative Processing of ChatGPT Feedback

The third innovative research topic is *collaborative processing of ChatGPT feedback*, which can be integrated into either collaborative writing (CW) tasks or individual writing tasks. In CW tasks, students work in pairs or small groups to complete a writing assignment and then jointly discuss ChatGPT feedback on their drafts, based on which they collaboratively incorporate ChatGPT feedback into the revision of their CW. In individual writing tasks, students first complete the task on their own, and then discuss ChatGPT-generated feedback with a partner before incorporating the ChatGPT feedback individually into their revisions.

This topic holds great promise, particularly given the well-documented benefits of collaboration in L2 writing, especially in CW contexts (Storch, 2013). Collaborative processing of feedback is also grounded in theoretical frameworks such as 1) Schmidt's Noticing Hypothesis (1990), which suggests that learning occurs when learners cognitively engage with input, and 2) Swain's Output Hypothesis (1985), which posits that learners' languaging, verbal or written deliberations on language-related episodes, improves the quality of outputs. Pedagogical support for collaborative feedback processing has been reported in previous research (e.g., Storch & Alshuraidah, 2020; Wigglesworth & Storch, 2012). For instance, in Storch and colleague's study, collaborative processing of teacher feedback in two writing tasks led to improved writing quality, including greater accuracy.

Extending the collaborative processing of teacher feedback, the author's research team (Taha and Li) examined undergraduate students' engagement in the collaborative processing of ChatGPT feedback and the impact of this collaboration on the qualities of their CW. The students in the First-Year-Writing class at a public university in the US worked in small groups to complete two argumentative essays, and then collaboratively discussed ChatGPT feedback on their drafts. The student

Table 6. Small group's collaborative processing of ChatGPT feedback

First draft
In conclusion, The advantages of becoming a bilingual in addition to speaking. From cognitive and social benefits to economic and academic advantages, being able to speak seconds language offer a range of positive outcomes. As our world is getting connected, the ability to communicate across languages and cultures will be a valuable skill. Enhancing both professional and personal lives.
ChatGPT feedback
The conclusion feels a little rushed and could benefit from a stronger restatement of the thesis. It is also important to avoid introducing new ideas in the conclusion (e.g., "in addition to speaking" seems like a new point that hasn't been fully developed in the essay). The last sentence could be strengthened by more directly linking back to the initial claim about the growing value of bilingualism in a globalized world.
Group discussion
Suraj: The previous argument, I just introduced a new topic in the conclusion. Jack: What did you say? I don't know. Skyler: You still got it there? Suraj: Yeah. Skyler: Oh, my gosh. Suraj: Yeah. Or just rewrite it. Skyler: But then you're talking about a whole new thing. How do you just rewrite a whole new thing? Jack: No. Suraj: Just fix all the mistakes in the essay. Skyler: What are you going to do with the conclusion? Suraj: We can just change it up a bit because for the most part it's perfect. Skyler: Let's restate the advantages of becoming bilingual in a globalized world.

received thorough training on CW and collaborative feedback processing, including how to seek ChatGPT feedback using specific prompts aligned with grading rubrics provided by their instructor.

The students were found to be highly engaged in discussing ChatGPT feedback in small groups, as illustrated in Table 6. The CW draft in question needs much improvement due to language errors and rhetorical issues. ChatGPT's feedback focuses on tightening up the theme of the paper and reinforcing the central claim. During the collaborative feedback processing, Skyler assumed the role of leader and reminded Suraj that ChatGPT had negatively commented on the introduction of new ideas in the conclusion. After learning Suraj's mere intention to fix language mistakes, Skyler pointed out the revising direction of restating the advantages of becoming bilingual in a globalized world, as suggested by ChatGPT.

Moreover, the small group incorporated the majority of ChatGPT's feedback into their revisions of CW, which enhanced the overall quality of their CW products. Table 7 below shows an example of students' incorporation of ChatGPT feedback. ChatGPT clearly pointed out two specific mistakes in the students' CW draft, and the small group successfully revised their work based on ChatGPT feedback.

In short, collaborative processing of ChatGPT feedback is a promising topic in L2 writing. Researchers can continue to explore this topic in diverse L2 writing contexts and writing tasks. Utilizing triangulated data sources, future research can examine students' and teachers' perspectives of collaborative processing of ChatGPT feedback for both IW tasks and CW tasks, and students' incorporation of ChatGPT feedback into revisions. Future research can also compare collaborative processing and individual processing of ChatGPT feedback, highlighting the role of collaboration in addressing this emerging L2 writing topic.

ChatGPT for L2 Writing Assessment

The final topic discussed in this paper is the potential of ChatGPT for L2 writing assessment, which has increasingly captured researchers' attention. ChatGPT has been found to provide reliable ratings and valuable feedback to test takers (Mizumoto & Eguchi, 2023), following similar rating

Table 7. Incorporation of ChatGPT feedback into revision

Excerpt in the original essay
Proficiency in a second language can offer more opportunities in every aspect and can create better productivity socially, economically, **and other cultural domains. Which makes** this ability a very desirable asset to have with you.
ChatGPT feedback
The phrase “and other cultural domains” needs better parallelism with the rest of the sentence. The use of “Which makes this ability” should be revised. Starting a sentence with “Which” creates a fragment. Instead, it can be combined with the first sentence or rephrased.
Revision
Proficiency in a second language can offer more opportunities in every aspect and can create better productivity socially, economically, and **in other cultural domains, which makes** this ability a very desirable asset to have with you.

patterns for low-, medium-, and high- quality essays, in alignment with TOEFL ratings (Mizumoto & Eguchi, 2023). Poole and Coss (2024) also maintained that ChatGPT can be a valuable tool in L2 writing assessment, provided that it is used strategically with well-crafted prompts. Several research studies have focused on ChatGPT's ability to assess writing accuracy. For instance, Jiang et al. (2023) reported that LLMs have great potential in supporting L2 writing accuracy assessment, with ChatGPT-4 excelling in precision. Moving beyond accuracy, future research should thoroughly explore ChatGPT's role in assessing multiple dimensions of L2 writing.

The author's research team (Gjorevski, Li, and Cox) conducted an empirical study to examine if ChatGPT can assess L2 writing like human raters, drawing on the NATO STANAG 6001 testing database[5]. 100 essays were randomly selected from Benchmark Advisory Test (BAT) 2, with 32 experienced human raters from twelve nations volunteering to assess these essays. ChatGPT was trained to assess the writing according to the same grading rubrics that human raters used. The human raters refreshed their knowledge by reviewing STANAG 6001 rating scale descriptions, supplemented with additional rating tools. Two raters, not involved in the rating process, provided written rationales on benchmark samples, which were used to train ChatGPT. ChatGPT was then assigned the role of a writing test rater and its task was explained through the prompt it received. After confirming its understanding of this role, the research team explained the writing tasks using the same samples, rating scale descriptors and other rating tools provided to the human raters. ChatGPT then graded the essays and completed justifications for their ratings, and the results were compared with those from the human raters.

The results revealed substantial agreement between GPT-3.5's ratings and human ratings. Given the variability in human ratings, the research team used Many-Facet Rasch Measurement (MFRM) and found that the human ratings were reliable when using the Fair Average. The correlation between ChatGPT and the Human MFRM Fair Average was significant (rho = .84, df = 98, p <. 001), indicating a strong correlation between ChatGPT and human ratings. A detailed textual analysis of all justifications was conducted to compare the justifications provided by human raters and ChatGPT and many similarities in the rationales were detected. For example, the two types of raters commented on the same aspects of writing, including comprehensibility, content development, task completion, cohesion and coherence, and lexical complexity and appropriateness. Interestingly, there were notable stylistic differences between the two, such as many instances of supportive comments from ChatGPT in contrast to few instances of human praises, and the human raters' direct and consistent use of phrases from the grading rubrics versus ChatGPT's indirect use of descriptors.

ChatGPT for L2 assessment will continue to be a hot research topic. Future research can examine ChatGPT as an assessment tool in diverse settings, including both classroom assessments and high-stake testing, where it can supplement human raters. Comparative studies could evaluate ChatGPT's performance (versions 3.5 and 4) in assessing L2 writing against other tools such as Baidu

AI Cloud and Grammarly. Also, more research could investigate different stakeholders' perceptions of ChatGPT as an automated writing evaluation (AWE) or automated essay scoring (AES) tool. Importantly, ChatGPT also holds potential for providing training materials for novice human raters after undergoing sufficient training itself. Future studies could explore ChatGPT's role as a potential assessment trainer. It is believed that collaboration between AI and human raters will significantly advance the field of L2 writing assessment.

CONCLUSION

This conceptual paper provides insights into leveraging ChatGPT for L2 writing feedback and assessment, highlighting four innovative research topics: ChatGPT-student collaboration during the writing process, ChatGPT-supported teacher feedback, collaborative processing of ChatGPT feedback, and using ChatGPT for writing assessment. ChatGPT assumes multiple roles in L2 writing, including that of a writing companion, teaching assistant, evaluator, and assessor. Although ChatGPT-3.5 (now called ChatGPT 4o) was used in the sample studies discussed in this paper, the four research topics can also inform inquiries involving other GenAI tools, such as Gemini, Wenxinyiyan and iFLYTEK. While its constraints such as false information and possible academic plagiarism (Yuan et al., 2024) are acknowledged, ChatGPT's benefits should be further explored for L2 pedagogy. To maximize the benefits of ChatGPT, teachers must receive thorough training on how to integrate AI into language instruction. The training should cover essential components, such as accessing ChatGPT, understanding the benefits and constraints of ChatGPT, prompt engineering, and using ChatGPT for lesson planning, task design, and assessment. With well-designed teacher training, more L2 practitioners will adopt ChatGPT judicially and effectively for L2 pedagogy and that more L2 researchers will further explore ChatGPT's potential for advancing L2 acquisition in the GenAI-mediated age.

CONFLICTS OF INTEREST

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REFERENCES

Aljasir, N., (2021). Matches or mismatches? Exploring shifts in individuals' beliefs about written corrective feedback as students and teachers-to-be. *Journal of Teaching and Teacher Education*, *9*(1), 1–10. DOI: 10.12785/jtte/090101

Bao, Y., & Li, B. (2023). A preliminary study on graduate student instructors' exploration, perception, and use of ChatGPT. *International Journal of Computer-Assisted Language Learning and Teaching*, *13*(1), 1–23. DOI: 10.4018/IJCALLT.332873

Barrot, J. S. (2023). Using ChatGPT for second language writing: Pitfalls and potentials. *Assessing Writing*, *57*, 100745. DOI: 10.1016/j.asw.2023.100745

Cheng, D., Li, M. Lee, T. (under review). Exploring the role of ChatGPT in ESL students' research writing. Manuscript submitted for publication.

Ferris, D. R. (2010). Second language writing research and written corrective feedback in SLA: Intersections and practical applications. *Studies in Second Language Acquisition*, *32*(2), 181–201. DOI: 10.1017/S0272263109990490

Fokides, E., & Peristeraki, E. (2024). Comparing ChatGPT's correction and feedback comments with that of educators in the context of primary students' short essays written in English and Greek. *Education and Information Technologies*, 1–45. DOI: 10.1007/s10639-024-12912-8

Geçkin, V., Kızıltaş, E., & Çınar, Ç. (2023). Assessing second-language academic writing: AI vs. Human raters. *Journal of Educational Technology and Online Learning*, *6*(4), 1096–1108. DOI: 10.31681/jetol.1336599

Gjorevski, A., Li, M., Cox, T. (under review). Exploring the potential of ChatGPT for evaluating English essays: Comparison with human raters. Manuscript submitted for publication.

Guo, K., & Wang, D. (2024). To resist it or to embrace it? Examining ChatGPT's potential to support teacher feedback in EFL writing. *Education and Information Technologies*, *29*(7), 8435–8463. DOI: 10.1007/s10639-023-12146-0

Han, J., & Li, M. (2024). Exploring ChatGPT-supported teacher feedback in the EFL context. *System*, *126*, 103502. DOI: 10.1016/j.system.2024.103502

Jiang, Z., Xu, Z., Pan, Z., He, J., & Xie, K. (2023). Exploring the role of artificial intelligence in facilitating assessment of writing performance in second language learning. *Languages (Basel, Switzerland)*, *8*(4), 247. DOI: 10.3390/languages8040247

Mahapatra, S. (2024). Impact of ChatGPT on ESL students' academic writing skills: A mixed methods intervention study. *Smart Learning Environments*, *11*(1), 9. DOI: 10.1186/s40561-024-00295-9

Mizumoto, A., & Eguchi, M. (2023). Exploring the potential of using an AI language model for automated essay scoring. *Research Methods in Applied Linguistics*, *2*(2), 100050. DOI: 10.1016/j.rmal.2023.100050

Mizumoto, A., Shintani, N., Sasaki, M., & Teng, M. F. (2024). Testing the viability of ChatGPT as a companion in L2 writing accuracy assessment. *Research Methods in Applied Linguistics*, *3*(2), 100116. DOI: 10.1016/j.rmal.2024.100116

Nkhobo, T., & Chaka, C. (2023). Student-Written versus ChatGPT-generated discursive essays: A comparative coh-metrix analysis of lexical diversity, syntactic complexity, and referential cohesion. *International Journal of Education and Development Using Information and Communication Technology*, *19*(3), 69–84. http://ijedict.dec.uwi.edu/viewarticle.php?id=3310.

Pfau, A., Polio, C., & Xu, Y. (2023). Exploring the potential of ChatGPT in assessing L2 writing accuracy for research purposes. *Research Methods in Applied Linguistics*, *2*(3), 100083. DOI: 10.1016/j.rmal.2023.100083

Poole, F. J., & Coss, M. D. (2024). Can ChatGPT Reliably and Accurately Apply a Rubric to L2 Writing Assessments? The Devil is in the Prompt (s). *Journal of Technology & Chinese Language Teaching*, *15*(1). https://doi.org/10.35542/osf.io/3r2zb

Ranade, N., & Eyman, D. (2024). Special Issue on Composing with Generative AI. *Computers and Composition*, *71*, 102834. DOI: 10.1016/j.compcom.2024.102834

Schmidt, R. W. (1990). The role of consciousness in second language learning. *Applied Linguistics*, *11*(2), 129–158. DOI: 10.1093/applin/11.2.129

Storch, N. (2013). *Collaborative writing in L2 classrooms*. Multilingual Matters. DOI: 10.21832/9781847699954

Storch, N., & Alshuraidah, A. (2020). Languaging when providing and processing peer feedback. In Storch, N., & Wataru, S. (Eds.), *Languaging in Language Teaching and Learning: A Collection of Empirical Studies* (pp. 149–169). John Benjamins. DOI: 10.1075/lllt.55.05sto

Strobl, C., Menke-Bazhutkina, I., Abel, N., & Michel, M. (2024). Adopting ChatGPT as a writing buddy in the advanced L2 writing class. *Technology in Language Teaching & Learning*, *6*(1), 1–19. DOI: 10.29140/tltl.v6n1.1168

Swain, M. (1985). Communicative competence: Some roles of comprehensible input and comprehensible output in its development. In Gass, S., & Madden, C. (Eds.), *Input in second language acquisition*. Newbury House.

Wang, C. (2024). Exploring students' generative AI-assisted writing processes: Perceptions and experiences from native and nonnative English speakers. Technology, Knowledge and Learning. https://doi.org/10.1007/s10758-024-09744-3

Warschauer, M., Tseng, W., Yim, S., Webster, T., Jacob, S., Du, Q., & Tate, T. (2023). The affordances and contradictions of AI-generated text for writers of English as a second or foreign language. *Journal of Second Language Writing*, *62*, 101071. DOI: 10.1016/j.jslw.2023.101071

Warschauer, M., & Xu, Y. (2024). Artificial intelligence for language learning: Entering a new era. *Language Learning & Technology*, *28*(2). https://hdl.handle.net/10125/73569

Wigglesworth, G., & Storch, N. (2012). Feedback and writing development through collaboration: A socio-cultural approach. In Manchón, R. (Ed.), *L2 writing development: Multiple perspectives* (pp. 69–100). De Gruyter Mouton. DOI: 10.1515/9781934078303.69

Yan, D. (2023). Impact of ChatGPT on learners in a L2 writing practicum: An exploratory investigation. *Education and Information Technologies*, *28*(11), 13943–13967. DOI: 10.1007/s10639-023-11742-4

Yuan, Y., Li, H., & Sawaengdist, A. (2024). The impact of ChatGPT on learners in English academic writing: Opportunities and challenges in education. *Language Learning in Higher Education*, *14*(1), 41–56. DOI: 10.1515/cercles-2023-0006

Zindela, N. (2023). Comparing measures of syntactic and lexical complexity in artificial intelligence and L2 human-generated argumentative essays. *International Journal of Education and Development Using Information and Communication Technology*, *19*(3), 50–68. http://ijedict.dec.uwi.edu/viewarticle.php?id=3312

Zou, M., & Huang, L. (2023). The impact of ChatGPT on L2 writing and expected responses: Voice from doctoral students. *Education and Information Technologies*, 1–19. https://doi.org/10.1007/s10639-023-12397-x